

Measure Theory – 4

The Lebesgue Integral, Convergence Theorems, Integration over Restricted Sets, Push-Forward Measures and Change of Variables, Differentiation Under the Integral Sign, and Tail Bounds

Ravleen Bajaj

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space, and in the probability setting $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space with $\mathbb{E}[Z] = \int_{\Omega} Z \, d\mathbb{P}$.

1 The Lebesgue Integral: Construction and Basic Properties

1.1 Definition via Simple Functions

Definition 1.1 (Simple function and its integral). A **simple function** is a measurable $s : X \rightarrow [0, \infty)$ taking finitely many values, $s = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathcal{M}$ disjoint. Its integral is

$$\int_X s \, d\mu = \sum_{i=1}^n a_i \mu(A_i) \quad (\text{with the convention } 0 \cdot \infty = 0).$$

Definition 1.2 (Integral of a measurable function). • For measurable $f \geq 0$: $\int_X f \, d\mu = \sup \left\{ \int_X s \, d\mu : 0 \leq s \leq f, s \text{ simple} \right\} \in [0, \infty]$.

- For general measurable f , write $f = f^+ - f^-$ with $f^{\pm} = \max(\pm f, 0)$. If at least one of $\int f^+, \int f^-$ is finite, set $\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu$.
- f is **integrable** ($f \in L^1(\mu)$) if $\int f^+ < \infty$ and $\int f^- < \infty$, equivalently $\int |f| \, d\mu < \infty$.

In the probability setting, $\mathbb{E}[X] = \int X \, d\mathbb{P}$.

Lemma 1.3 (Approximation by simple functions). *If $f \geq 0$ is measurable, then*

$$s_k = \sum_{j=0}^{k2^k-1} \frac{j}{2^k} \mathbf{1}_{\{j2^{-k} \leq f < (j+1)2^{-k}\}} + k \mathbf{1}_{\{f \geq k\}}$$

are simple functions with $0 \leq s_1 \leq s_2 \leq \dots$ and $s_k \uparrow f$ pointwise. If f is bounded, the convergence is uniform.

This lemma is what makes the “ladder” proof strategy of Sections 3 and 4 work.

1.2 Basic Properties

Proposition 1.4 (Properties of the integral). *Let f, g be measurable and integrable (or non-negative where indicated).*

- (a) **Monotonicity:** $f \leq g \implies \int f \leq \int g$.
- (b) **Linearity:** $\int(\alpha f + \beta g) = \alpha \int f + \beta \int g$.
- (c) **Triangle inequality:** $|\int f| \leq \int |f|$.
- (d) **Null sets:** if $f \geq 0$, then $\int f \, d\mu = 0 \iff f = 0 \text{ } \mu\text{-a.e.}$
- (e) **A.e. equality:** $f = g \text{ a.e.} \implies \int f = \int g$.
- (f) **Finiteness:** $f \in L^1(\mu) \implies |f| < \infty \text{ a.e.}$

Proof. (a)–(c) follow from the definition through simple functions and f^+, f^- . For (d): if $f \geq 0$ and $\int f = 0$, then for each n , $\frac{1}{n} \mathbf{1}_{\{f \geq 1/n\}} \leq f$, so $\mu(f \geq \frac{1}{n}) \leq n \int f = 0$, and $\{f > 0\} = \bigcup_n \{f \geq \frac{1}{n}\}$ is a countable union of null sets. The converse holds because $f = 0$ a.e. forces every simple minorant to vanish a.e. (e) follows from (d) applied to $|f - g|$. For (f), if $\mu(|f| = \infty) > 0$ then $\int |f| \geq n \mu(|f| = \infty) \rightarrow \infty$ for every n . $\square$

Note:

- $\int f$ need not exist even for a nice-looking f : if both $\int f^+$ and $\int f^-$ are infinite, the integral is undefined ($\infty - \infty$).
- $f \geq 0$ and $\int f = 0$ does *not* mean $f \equiv 0$, only $f = 0$ a.e.

2 Classical Convergence Theorems

2.1 Monotone Convergence, Fatou, and Dominated Convergence

Theorem 2.1 (Monotone Convergence Theorem (MCT)). *If $0 \leq f_n \uparrow f$ μ -a.e., then $\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu$. In probability language: $0 \leq X_n \uparrow X$ a.s. implies $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$.*

Proof. By monotonicity $\int f_n$ increases to some $L \leq \int f$. For the reverse, fix a simple $0 \leq s \leq f$ and $c \in (0, 1)$, and set $E_n = \{f_n \geq cs\}$. Then $E_n \uparrow X$ (up to a null set), and

$$\int f_n \geq \int_{E_n} f_n \geq c \int_{E_n} s \, d\mu \xrightarrow{n \rightarrow \infty} c \int s \, d\mu,$$

because $E \mapsto \int_E s \, d\mu = \sum a_i \mu(A_i \cap E)$ is a measure and so is continuous from below. Hence $L \geq c \int s$; let $c \uparrow 1$ and take the supremum over s . $\square$

Theorem 2.2 (Fatou's Lemma). *If $f_n \geq 0$ are measurable, then*

$$\int \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int f_n \, d\mu.$$

In probability language: $X_n \geq 0 \implies \mathbb{E}[\liminf X_n] \leq \liminf \mathbb{E}[X_n]$.

Proof. Let $g_k = \inf_{n \geq k} f_n$. Then $0 \leq g_k \uparrow \liminf f_n$ and $g_k \leq f_n$ for every $n \geq k$, so $\int g_k \leq \inf_{n \geq k} \int f_n$. By MCT, $\int \liminf f_n = \lim_k \int g_k \leq \lim_k \inf_{n \geq k} \int f_n = \liminf_n \int f_n$. $\square$

Theorem 2.3 (Dominated Convergence Theorem (DCT)). *If $f_n \rightarrow f$ μ -a.e. and $|f_n| \leq g$ a.e. for some integrable g , then $f \in L^1(\mu)$ and*

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu, \quad \text{indeed} \quad \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

In probability language: $X_n \rightarrow X$ a.s. and $|X_n| \leq Y$ with $\mathbb{E}[Y] < \infty$ implies $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$.

Proof. Since $|f| \leq g$ a.e., f is integrable. Apply Fatou to $g + f_n \geq 0$ and to $g - f_n \geq 0$:

$$\begin{aligned} \int g + \int f &= \int \liminf (g + f_n) \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n, \\ \int g - \int f &= \int \liminf (g - f_n) \leq \liminf \int (g - f_n) = \int g - \limsup \int f_n. \end{aligned}$$

Since $\int g < \infty$ it may be cancelled, giving $\limsup \int f_n \leq \int f \leq \liminf \int f_n$. For the second statement apply this to $|f_n - f| \leq 2g$, which tends to 0 a.e. $\square$

Which theorem to use.

- **MCT:** $f_n \geq 0$ and *increasing*; no integrability needed, the limit may be $+\infty$.
- **Fatou:** $f_n \geq 0$ only; gives an *inequality*, never an equality in general.
- **DCT:** $f_n \rightarrow f$ with a *single integrable dominating function* g that does not depend on n .

2.2 Consequences and Counterexamples

Corollary 2.4 (Termwise integration of series). (a) *If $f_n \geq 0$, then $\int \sum_n f_n d\mu = \sum_n \int f_n d\mu$ (MCT applied to partial sums).*

(b) *If $\sum_n \int |f_n| d\mu < \infty$, then $\sum_n f_n$ converges absolutely a.e., and $\int \sum_n f_n = \sum_n \int f_n$ (DCT with $g = \sum_n |f_n|$, which is integrable by part (a)).*

Corollary 2.5 (Bounded Convergence Theorem). *If $\mu(X) < \infty$, $f_n \rightarrow f$ a.e. and $|f_n| \leq K$, then $\int f_n \rightarrow \int f$.*

Example 2.6 (Fatou can be strict; DCT can fail). On $((0, 1), \mathcal{B}, \lambda)$ let $f_n = n \mathbf{1}_{(0, 1/n)}$. Then $f_n \rightarrow 0$ pointwise, yet $\int f_n = 1$ for every n . So

$$0 = \int \liminf f_n < \liminf \int f_n = 1.$$

The mass “escapes to the boundary”, and no integrable g dominates all f_n (the smallest candidate is $\sup_n f_n \approx \lfloor 1/x \rfloor$, whose integral diverges). A second standard example on $\mathbb{R}$ is $f_n = \mathbf{1}_{[n, n+1]}$, where the mass escapes to infinity.

2.3 Problem: A Uniform Second-Moment Bound Passes to the Limit

Proposition 2.7. *Let $X_i \rightarrow X$ almost surely and suppose $\sup_i \mathbb{E}[X_i^2] \leq M < \infty$. Then*

- (a) $\mathbb{E}[X^2] \leq M < \infty$;
- (b) $\mathbb{E}|X_i - X| \rightarrow 0$ (hence $\mathbb{E}[X_i] \rightarrow \mathbb{E}[X]$).

Proof. (a) Put $Y_i = X_i^2 \geq 0$. Then $Y_i \rightarrow X^2$ a.s., so by Fatou

$$\mathbb{E}[X^2] = \mathbb{E}\left[\lim_i X_i^2\right] \leq \liminf_{i \rightarrow \infty} \mathbb{E}[X_i^2] \leq M.$$

(b) Note that no dominating function is available, so DCT does not apply directly. Truncate: let $\varphi_K(x) = \max(-K, \min(x, K))$. Then $|x - \varphi_K(x)| \leq |x| \mathbf{1}_{\{|x| > K\}} \leq x^2/K$, hence

$$\mathbb{E}|X_i - \varphi_K(X_i)| \leq \frac{M}{K}, \quad \mathbb{E}|X - \varphi_K(X)| \leq \frac{M}{K}.$$

Since φ_K is continuous and bounded by K , $\varphi_K(X_i) \rightarrow \varphi_K(X)$ a.s. and DCT (or bounded convergence) gives $\mathbb{E}|\varphi_K(X_i) - \varphi_K(X)| \rightarrow 0$. By the triangle inequality, $\limsup_i \mathbb{E}|X_i - X| \leq 2M/K$ for every K , and letting $K \rightarrow \infty$ gives the claim. $\square$

Remark. The uniform bound $\sup_i \mathbb{E}[X_i^2] < \infty$ is a special case of *uniform integrability*: it forces $\sup_i \mathbb{E}[|X_i| \mathbf{1}_{\{|X_i| > K\}}] \leq M/K \rightarrow 0$. Uniform integrability plus a.s. convergence is exactly what is needed for L^1 convergence, which is the correct replacement for DCT when no single dominating function exists. The same argument works for any bound $\sup_i \mathbb{E}|X_i|^p < \infty$ with $p > 1$.

Note:

- Pointwise convergence alone never justifies $\lim \int f_n = \int \lim f_n$ (Example above).
- DCT needs one g that works for *all* n ; checking $|f_n| \leq g_n$ with g_n integrable but varying with n is not enough.
- Fatou's Lemma requires $f_n \geq 0$ (or $f_n \geq -g$ with g integrable).

3 Integration Over Restricted Measurable Sets

3.1 Restricted Measure Space

Definition 3.1 (Integral over a set; restricted measure). Let $B \in \mathcal{M}$. The **integral of f over B** is $\int_B f \, d\mu := \int_X f \mathbf{1}_B \, d\mu$. The **restricted measure** μ_B is defined by

$$\mu_B(A) = \mu(A \cap B), \quad A \in \mathcal{M}.$$

Lemma 3.2. μ_B is a measure on $(X, \mathcal{M})$, concentrated on B (that is, $\mu_B(B^c) = 0$).

Proof. $\mu_B(\emptyset) = 0$, and for disjoint A_i the sets $A_i \cap B$ are disjoint with $\bigcup_i (A_i \cap B) = (\bigcup_i A_i) \cap B$, so σ -additivity is inherited from μ . $\square$

Theorem 3.3 (Integration over B equals integration against μ_B). *For any $\mathcal{M}$ -measurable function f ,*

$$\int_B f \, d\mu = \int_X f \, d\mu_B,$$

in the sense that one side exists if and only if the other does, and then they are equal.

Proof (the ladder procedure). The proof climbs from indicators to general functions.

1. **Indicators.** For $f = \mathbf{1}_A$: $\int_B \mathbf{1}_A \, d\mu = \int \mathbf{1}_{A \cap B} \, d\mu = \mu(A \cap B) = \mu_B(A) = \int_X \mathbf{1}_A \, d\mu_B$.

2. **Simple functions.** Let $s = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$. Then

$$\int_B s \, d\mu = \sum_{i=1}^n a_i \mu(A_i \cap B) \quad \text{and} \quad \int_X s \, d\mu_B = \sum_{i=1}^n a_i \mu_B(A_i) = \sum_{i=1}^n a_i \mu(A_i \cap B).$$

3. **Non-negative functions.** Let $f \geq 0$ and choose simple $s_k \uparrow f$ (Lemma 1.3). Then $s_k \mathbf{1}_B \uparrow f \mathbf{1}_B$, so by MCT under μ and then under μ_B ,

$$\int_B f \, d\mu = \lim_{k \rightarrow \infty} \int_B s_k \, d\mu = \lim_{k \rightarrow \infty} \int_X s_k \, d\mu_B = \int_X f \, d\mu_B.$$

4. **General measurable functions.** Write $f = f^+ - f^-$. Step 3 applies to f^+ and f^- separately, and $(f \mathbf{1}_B)^\pm = f^\pm \mathbf{1}_B$. Thus $\int_B f^\pm \, d\mu = \int_X f^\pm \, d\mu_B$; in particular $f \mathbf{1}_B$ is μ -integrable iff f is μ_B -integrable, and subtracting gives the identity. $\square$

3.2 Consequences

Proposition 3.4. *Let $f \in L^1(\mu)$ and define $\nu(B) = \int_B f \, d\mu$.*

- (a) **Additivity over the domain:** *for disjoint $B_1, B_2 \in \mathcal{M}$, $\int_{B_1 \cup B_2} f = \int_{B_1} f + \int_{B_2} f$; more generally ν is σ -additive (a signed measure), and a genuine measure when $f \geq 0$.*
- (b) **Null sets:** $\mu(B) = 0 \implies \nu(B) = 0$ (so $\nu \ll \mu$).
- (c) **Bounds:** $|\int_B f \, d\mu| \leq \int_B |f| \, d\mu \leq \int_X |f| \, d\mu$.

Proof. (a) $\mathbf{1}_{B_1 \cup B_2} = \mathbf{1}_{B_1} + \mathbf{1}_{B_2}$, and countable additivity follows by DCT with dominating function $|f|$ applied to partial sums of $\mathbf{1}_{B_i} f$. (b) $f \mathbf{1}_B = 0$ a.e. when $\mu(B) = 0$. (c) Triangle inequality and monotonicity. $\square$

Example 3.5. Let $\mu = \lambda|_{(0,1)} + 2\delta_{1/2}$, $f(x) = x^2$ and $B = [0, \frac{1}{2}]$. Using $\int_B f \, d\mu = \int f \, d\mu_B$ with $\mu_B = \lambda|_{(0,1/2]} + 2\delta_{1/2}$,

$$\int_B f \, d\mu = \int_0^{1/2} x^2 \, dx + 2 \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{24} + \frac{1}{2} = \frac{13}{24}.$$

Here the atom at $\frac{1}{2}$ contributes because $\frac{1}{2} \in B$ (the interval is closed).

Note:

- $\int_B f \, d\mu$ depends on whether endpoints belong to B exactly when μ has an atom there (as in the example).
- μ_B is a measure on all of X , not on B alone; it just puts no mass outside B .

4 Push-Forward Measures and Change of Variables

4.1 Induced Measures

Definition 4.1 (Push-forward measure). Let $f : (\Omega, \mathcal{F}, \mu_2) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable. The **push-forward** (image, or induced) measure of μ_2 under f is

$$\mu_1(A) = \mu_2(f^{-1}(A)) = \mu_2 \circ f^{-1}(A), \quad A \in \mathcal{B}(\mathbb{R}).$$

When $\mu_2 = \mathbb{P}$ and $f = X$ is a random variable, $\mu_1 = \mathbb{P}_X$ is the **law** (distribution) of X .

Lemma 4.2. $\mu_1 = \mu_2 \circ f^{-1}$ is a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Proof. $\mu_1(\emptyset) = \mu_2(\emptyset) = 0$. Preimages preserve disjointness and unions: if A_i are disjoint then so are $f^{-1}(A_i)$, and $f^{-1}(\bigcup_i A_i) = \bigcup_i f^{-1}(A_i)$, so σ -additivity transfers from μ_2 . $\square$

4.2 The Change-of-Variables Formula

Theorem 4.3 (Change of variables). Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be Borel measurable. Then

$$\int_{\mathbb{R}} g \, d\mu_1 = \int_{\Omega} (g \circ f) \, d\mu_2,$$

whenever $g \geq 0$, or $g \in L^1(\mu_1)$; in the latter case $g \circ f \in L^1(\mu_2)$ (and conversely).

Proof (the ladder procedure). 1. **Indicators.** For $g = \mathbf{1}_A$ with A Borel, $g \circ f = \mathbf{1}_{f^{-1}(A)}$, so

$$\int \mathbf{1}_A \, d\mu_1 = \mu_1(A) = \mu_2(f^{-1}(A)) = \int \mathbf{1}_{f^{-1}(A)} \, d\mu_2 = \int (\mathbf{1}_A \circ f) \, d\mu_2.$$

2. **Simple functions** follow by linearity.

3. **Non-negative g .** Take simple $s_k \uparrow g$; then $s_k \circ f \uparrow g \circ f$, and MCT on both sides gives the identity.

4. **General g .** Apply step 3 to $g^\pm$ (note $(g \circ f)^\pm = g^\pm \circ f$); integrability on either side follows from the identity for $|g|$, and subtracting finishes the proof. $\square$

Special cases.

- **Mean.** Setting $g(w) = w$:

$$\int_{\mathbb{R}} w \, d\mu_1(w) = \int_{\Omega} f(\omega) \, d\mu_2(\omega).$$

For $\mu_2 = \mathbb{P}$ and $f = X$ this is $\mathbb{E}[X] = \int w \, d\mathbb{P}_X(w)$: the expectation depends only on the law of X .

- **Law of the unconscious statistician.** For Borel g , $\mathbb{E}[g(X)] = \int g \, d\mathbb{P}_X$, with $= \int g(w)p(w) \, dw$ if $\mathbb{P}_X$ has density p and $= \sum_w g(w)\mathbb{P}(X = w)$ if $\mathbb{P}_X$ is discrete.

- **Distribution functions.** If F is the distribution function of $\mathbb{P}_X$, then $\mathbb{E}[g(X)] = \int g \, d\mu_F$ (Lebesgue–Stieltjes integral).

Example 4.4. Let $\Omega = (0, 1)$ with Lebesgue measure and $f(\omega) = -\ln \omega$. For $a \geq 0$,

$$\mu_1((a, \infty)) = \lambda(\{\omega : -\ln \omega > a\}) = \lambda((0, e^{-a})) = e^{-a},$$

so μ_1 is the $\text{Exp}(1)$ law. Both sides of the change-of-variables formula with $g(w) = w$ agree:

$$\int_0^1 (-\ln \omega) \, d\omega = 1 = \int_0^\infty w e^{-w} \, dw.$$

Note:

- g must be *Borel* measurable and f measurable, so that $g \circ f$ is measurable.
- The push-forward lives on the *target* space; $\mu_1(\mathbb{R}) = \mu_2(\Omega)$, so total mass is preserved.
- f need not be injective: the formula holds with no Jacobian, since it is a statement about measures, not about substitution in Riemann integrals.

5 Differentiation Under the Integral Sign

5.1 The Leibniz Integral Rule

Theorem 5.1 (Leibniz rule). *Let $F(t) = \int_X f(x, t) \, d\mu(x)$ for $t \in (a, b)$. Suppose*

- (i) $x \mapsto f(x, t)$ is integrable for every $t \in (a, b)$;
- (ii) for μ -almost every x , $t \mapsto f(x, t)$ is differentiable on (a, b) , with derivative $\frac{\partial f}{\partial t}(x, t)$;
- (iii) there is a μ -integrable g , not depending on t , with

$$\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x) \quad \text{for all } t \in (a, b) \text{ and a.e. } x.$$

Then F is differentiable on (a, b) and

$$F'(t) = \int_X \frac{\partial f}{\partial t}(x, t) \, d\mu(x).$$

Proof. Fix $t \in (a, b)$ and any sequence $h_n \rightarrow 0$, $h_n \neq 0$, with $t + h_n \in (a, b)$. Define

$$D_n(x) = \frac{f(x, t + h_n) - f(x, t)}{h_n}.$$

Each D_n is measurable, and $D_n(x) \rightarrow \frac{\partial f}{\partial t}(x, t)$ for a.e. x (which shows this limit is measurable). By the Mean Value Theorem there is $\xi_n(x)$ between t and $t + h_n$ with $D_n(x) = \frac{\partial f}{\partial t}(x, \xi_n(x))$, so $|D_n(x)| \leq g(x)$. DCT gives

$$\frac{F(t + h_n) - F(t)}{h_n} = \int D_n \, d\mu \longrightarrow \int \frac{\partial f}{\partial t}(x, t) \, d\mu(x).$$

The sequence (h_n) was arbitrary, so $F'(t)$ exists and equals this integral. □

Proposition 5.2 (Continuity under the integral sign). *If $f(\cdot, t)$ is integrable for each t , $t \mapsto f(x, t)$ is continuous at t_0 for a.e. x , and $|f(x, t)| \leq g(x)$ for an integrable g and all t near t_0 , then F is continuous at t_0 .*

Proof. For any $t_n \rightarrow t_0$, $f(x, t_n) \rightarrow f(x, t_0)$ a.e. and $|f(x, t_n)| \leq g(x)$, so DCT applies. $\square$

Example 5.3. For $t > 0$ let $F(t) = \int_0^\infty e^{-tx} dx = \frac{1}{t}$. Fix $t_0 > 0$ and work on $t \in (t_0, \infty)$:

$$\left| \frac{\partial}{\partial t} e^{-tx} \right| = x e^{-tx} \leq x e^{-t_0 x} =: g(x),$$

and g is integrable on $(0, \infty)$. Theorem 5.1 gives

$$F'(t) = - \int_0^\infty x e^{-tx} dx = -\frac{1}{t^2},$$

which agrees with differentiating $1/t$ directly. Since $t_0 > 0$ was arbitrary, the formula holds for every $t > 0$, although no single dominating function works on all of $(0, \infty)$.

Remark. • Differentiability is a *local* property, so it suffices to find a dominating function on a neighbourhood of each t (as in the example), not one on the whole interval.

- Repeating the argument (with $\partial^k f / \partial t^k$ dominated) gives differentiation of any order; this justifies, for instance, $\frac{d^k}{dt^k} \mathbb{E}[e^{tX}] = \mathbb{E}[X^k e^{tX}]$ inside the interval where the moment generating function is finite.
- For μ counting measure this is the theorem on termwise differentiation of series.

Note:

- The bound in (iii) must be uniform in t and integrable in x ; a bound like $|\partial_t f(x, t)| \leq C(t)$ with C depending on t is not enough.
- Domination is on the *derivative*, not on f itself.
- Without domination, $\frac{d}{dt} \int f \neq \int \partial_t f$ can occur (mass may escape as in the Fatou example of Section 2).

6 Tail-Bound Inequalities

6.1 Markov and Chebyshev

Theorem 6.1 (Markov's inequality). *If $Y \geq 0$ and $k > 0$, then $\mathbb{P}(Y \geq k) \leq \frac{\mathbb{E}[Y]}{k}$. More generally, if $\varphi \geq 0$ is non-decreasing on the range of Y and $\varphi(k) > 0$, then $\mathbb{P}(Y \geq k) \leq \mathbb{E}[\varphi(Y)] / \varphi(k)$.*

Proof. $k \mathbf{1}_{\{Y \geq k\}} \leq Y$ pointwise. Taking expectations and using monotonicity of the integral gives $k \mathbb{P}(Y \geq k) \leq \mathbb{E}[Y]$. The general case applies this to $\varphi(Y)$. $\square$

Corollary 6.2 (Chebyshev's inequality). *If X has mean μ and variance $\sigma^2 < \infty$, then for $a > 0$,*

$$\mathbb{P}(|X - \mu| \geq a\sigma) \leq \frac{1}{a^2}.$$

Proof. Apply Markov to $Y = (X - \mu)^2$ with $k = a^2 \sigma^2$. $\square$

6.2 One-Sided Bound (Cantelli)

Theorem 6.3 (Cantelli's inequality). *Let X have mean μ and variance $\sigma^2 \in (0, \infty)$. For $a > 0$,*

$$\mathbb{P}(X \geq \mu + a\sigma) \leq \frac{1}{1 + a^2}.$$

Equivalently, with $t = a\sigma$, $\mathbb{P}(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}$.

Proof. Consider the non-negative random variable

$$Y = ((X - \mu)a + \sigma)^2 \geq 0.$$

1. **Expectation of Y .** Expanding and using $\mathbb{E}[X - \mu] = 0$, $\mathbb{E}[(X - \mu)^2] = \sigma^2$:

$$\mathbb{E}[Y] = a^2 \mathbb{E}[(X - \mu)^2] + 2a\sigma \mathbb{E}[X - \mu] + \sigma^2 = a^2 \sigma^2 + 0 + \sigma^2 = \sigma^2(1 + a^2).$$

2. **Inclusion of events.** If $X \geq \mu + a\sigma$, then $(X - \mu)a \geq a^2\sigma$, so $(X - \mu)a + \sigma \geq \sigma(1 + a^2) > 0$. Squaring (both sides non-negative), $Y \geq \sigma^2(1 + a^2)^2$. Hence $\{X \geq \mu + a\sigma\} \subseteq \{Y \geq \sigma^2(1 + a^2)^2\}$.
3. **Apply Markov.** With $k = \sigma^2(1 + a^2)^2$,

$$\mathbb{P}(X \geq \mu + a\sigma) \leq \mathbb{P}(Y \geq \sigma^2(1 + a^2)^2) \leq \frac{\sigma^2(1 + a^2)}{\sigma^2(1 + a^2)^2} = \frac{1}{1 + a^2}. \quad \square$$

Remark (Where the choice of Y comes from). For any $u \geq 0$, the same argument with $Y_u = (X - \mu + u)^2$ gives

$$\mathbb{P}(X - \mu \geq t) \leq \frac{\mathbb{E}[Y_u]}{(t + u)^2} = \frac{\sigma^2 + u^2}{(t + u)^2}.$$

The right side is minimized at $u = \sigma^2/t$, giving $\sigma^2/(\sigma^2 + t^2)$. With $t = a\sigma$ this is $u = \sigma/a$, and $Y_u = Y/a^2$, which is exactly the variable used above.

6.3 Comparison, Sharpness, and Variants

- **Better than Chebyshev for one tail.** Since $\frac{1}{1+a^2} < \frac{1}{a^2}$, Cantelli improves on the Chebyshev bound $1/a^2$ (which controls both tails at once). For instance, at $a = 2$: Chebyshev gives 0.25, Cantelli gives 0.2.
- **Lower tail.** Applying the theorem to $-X$, $\mathbb{P}(X \leq \mu - a\sigma) \leq \frac{1}{1+a^2}$.
- **Sharpness.** Let $X = \mu + a\sigma$ with probability $\frac{1}{1+a^2}$ and $X = \mu - \sigma/a$ with probability $\frac{a^2}{1+a^2}$. Then $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$, and $\mathbb{P}(X \geq \mu + a\sigma) = \frac{1}{1+a^2}$, so the bound cannot be improved using only the mean and variance.
- **Two-sided use.** Adding the two tails gives $\mathbb{P}(|X - \mu| \geq a\sigma) \leq \frac{2}{1+a^2}$, which is weaker than Chebyshev once $a > 1$; use Chebyshev for two-sided bounds.

Note:

- Markov requires $Y \geq 0$; in Cantelli's proof this is why one squares the shifted variable.

- The step “square both sides” in the inclusion needs both sides non-negative, which is where $a > 0$ and $\sigma > 0$ are used.
- If $\sigma = 0$ then $X = \mu$ a.s. and the bound is trivial.
- These inequalities only use the first two moments; for light-tailed variables exponential (Chernoff-type) bounds via $\varphi(y) = e^{\theta y}$ in Markov’s inequality are much stronger.

References

- [1] S. Basu, T. Butler, D. Estep, and N. Panda, *A Ramble Through Probability: How I Learned to Stop Worrying and Love Measure Theory*, Computational Science & Engineering, vol. 29, SIAM, Philadelphia, 2024.